Rohan Mehra

Graduate Student in Computer Science

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Education

2025 – present **MS in Computer Science**, *ISIMA, Clermont Auvergne INP, UCA*
- Ranked: 8/34
- *Selected Courses:* Advanced Machine Learning and Data Mining, High Performance Computing, Decision and Learning under Uncertainty.

2021 – 2025 **BS in Electrical Engineering and Computer Science**, *IISER Bhopal*
- Grade: 8.92/10
- *Selected Courses:* Data Science and Machine Learning, Multi-agent Reinforcement Learning, Algorithms, Operating System, Computer Organization, Theory of Computation, Probability and Statistics, Linear Algebra, Competitive Programming.

Research Experience

Mar. – present 2026 **LIFAT, CNRS, Université de Tours, France**, *M2 Research Intern*
- Advised by Prof. Gilles Venturini as part of RFAI team.
- Working on AI-assisted image recognition of bumblebees *(bombus)*.
- Using transfer learning, and explainable AI approaches.

May 2025 – Mar. 2026 & May – July 2024 **Institut Pascal, CNRS, Université Clermont Auvergne, France**, *Research Intern*
- Advised by Dr. Mathieu Labussière as part of ComSee team.
- Worked on deep learning models on SWIR images' for road perception in adverse weather conditions.
- Models worked on include Grounded SAM, YOLO, RF-DETR [Sem. Segmentation] and CUT, CycleGAN, Pix2Pix, BBDM [GANs & diffusion-based]

Jan. – April 2025 **IISER Bhopal, India**, *BS Project Student*
- Advised by Dr. Arpit Sharma at the Dept. of EECS.
- Worked on using Zero-Shot Learning model for Sector Classification, utilizing the CrowdRE dataset.
- Used models including Sentence-BERT, All-MiniLM-L12, Bert4RE and BERTOverflow

May – July 2023 **Indian Institute of Space Science and Technology**, *Research Intern*
- Advised by Prof. Deepak Mishra as part of the CVVR Lab
- Worked on robotics and implementation of SLAM through Kalman filter and EFK Methods.
- Used Robot Operating System (ROS), RViz, and Gazebo.

Selected Publications

Full list available on Google Scholar

Rohan Mehra, Alexandre Riffard, Yannis Loumouamou, Mathieu Labussière, "Analysis of SWIR Imaging Detection Performance Under Adverse Environmental Conditions for Autonomous Driving Systems", ***Under review*** *at IEEE Transactions on Intelligent Vehicles (**T-IV**), 2026.*

Yannis Loumouamou, Alexandre Riffard, **Rohan Mehra**, Mathieu Labussière, "Experimental study of YOLOv8 fine-tuning for RGB-SWIR multimodal road detection", ***Under review*** *at Reconnaissance des Formes, Image, Apprentissage et Perception (**RFIAP**), 2026.*

Rohan Mehra, Alexandre Riffard, Mathieu Labussière, Pierre Duthon, Romuald Aufrère, "Would SWIR modality help for detection and segmentation in harsh weather conditions? An experimental study.", *Proceedings of the IEEE/CVF International Conference on Computer Vision Workshops (**ICCVW**), 2025.* [Oral]

Teaching Experience

2024 & 2023 **Basic Electronics**, *IISER Bhopal* ($\times$ 2)

- Teaching Assistant for an undergraduate class of 180 students for 2024-25/23-24 Semester I.
- Conducted tutorial sessions, prepared assignment solutions, and provided one-to-one guidance.

Academic / Course Projects

2025 **Risk-Aware Multi-Epoch Package Delivery Planning**, IISER Bhopal

- Used greedy algorithm to solve package delivery problems while balancing risk and reward.
- Developed a GUI-based visualization tool using Tkinter and Matplotlib.

2024 **Training Multi-Robot Warehouse (RWARE) Environment**, IISER Bhopal

- Trained robots for optimized coordination in the RWARE environment.
- Employed Deep Q-Network (DQN) for efficient learning.

2023 **Prediction of Mortality Rate of Heart Failure Patients in ICU**, IISER Bhopal

- Built a classification model to predict ICU mortality in heart failure patients.
- Utilized Random Forest, Logistic Regression, and AdaBoost etc. for model development.

Technical Skills

Languages Python (with PyTorch framework), C++, C, Java

Software ROS, Gazebo, MATLAB, Git, LaTeX, Wolfram Mathematica

Events / Fellowships / Achievements

2026 **ELLIS Winter School on Foundation Models (FoMo)**

Selected to attend the FoMo Winter School hosted by the ELLIS unit Amsterdam, among a global pool of applicants ($\sim$200 participants).

2025 **MIAI Cluster Excellence Scholarship**

Selected for Excellence Scholarship program offered by the Multidisciplinary Institute in Artificial Intelligence (MIAI) in France to pursue master's program, awarded to $\sim$20 candidates university wide.

2024 **IASc-INSA-NASI Summer Research Fellowship** *[Declined]*

Selected for summer research fellowship jointly offered by the Indian Academy of Sciences, Indian National Science Academy, and National Academy of Sciences, awarded to the top $\sim$5% of applicants.

2021 **Indian Maritime University Undergraduate Admissions**

Secured All India Rank 9 out of 14,108 ranked candidates, placing in the top 0.1% nationwide.

External / Online Courses

- CS231n: Deep Learning for Computer Vision - (Offered by Stanford Online)
- Introduction to Deep Learning & Neural Networks with Keras – Coursera (Offered by IBM)
- Image Processing Onramp, Simulink Onramp, MATLAB Onramp – MathWorks

Positions of Responsibility / Extra Curriculars

2023 – 2024 **Cultural Secretary**, *IISER Bhopal*

Elected representative of the student body for organizing all cultural events at IISER Bhopal. Led a team of 70 performers to the the Inter IISER Cultural Meet 2023 held at IISER Mohali.

2022 **NCC 'B' Certificate**, *National Cadet Corps*, India

Completed 'B' division NCC military training from 8 UP BN NCC, Aligarh.

Other

Languages English (fluent, TOEFL 99/120), Hindi (Native), French (Basic)

Hobbies Table Tennis, Quizzing